

Automated Assessment of Cookie Consent Dialogues for GDPR Compliance

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Abstract—The General Data Protection Regulation (GDPR) mandates that user consent for data processing must be freely given, specific, informed, and unambiguous. But a lot of cookie consent dialogues on websites don't meet these standards. They use "dark patterns" to push users towards choices that aren't as private. This paper proposes a new, automated way to check if cookie consent dialogues follow the GDPR rules on a large scale. Our methodology combines web scraping techniques to extract consent dialogues, natural language processing (NLP) with a BERT-based model to analyze the textual content, and structural analysis to detect manipulative design patterns. We present a multi-faceted compliance score that rates banners based on important rules from the GDPR, the ePrivacy Directive, and the EDPB guidelines. To facilitate machine-readable compliance reporting, we standardize consent metadata using the Data Privacy Vocabulary (DPV-2). We have used our crawler on a dataset of more than 300 websites, and it found that many of them were not following the rules and that dark patterns like pre-ticked boxes and unfair choice architecture were common. Our NLP classifier gets an F1 score of 0.97 when it finds these patterns, which shows that automated, large-scale compliance auditing is possible. This study shows the regulators, businesses, and end users a way to improve transparency and enforce data protection standards across the web that can be used by everyone.

Index Terms—GDPR, Cookie Consent, Dark Patterns, Natural Language Processing, Web Scraping, Compliance Automation, DPV-2.

I. INTRODUCTION

The General Data Protection Regulation (GDPR) is a major step forward in data privacy law. It gives people in the European Union a lot of power over their own personal data. Article 4(11) [1] of the GDPR says that valid consent must be given freely, be clear, and be specific. This is one of the main parts of the GDPR. Most websites meet this requirement by having a cookie consent banner, which is the main way for users to manage their data preferences.

Even though the rule is very clear, there are a lot of implementations in the digital world that doesn't follow them. A lot of companies use "dark patterns" [2] to make consent dialogues on purpose that are difficult to understand, misleading, or forceful. These manipulative designs use cognitive biases to push the users into agreeing to data processing activities which they might not have agreed to otherwise. This goes against the very idea of informed consent. Because the web is so big, regulators can't manually check these banners, and

organisations that want to follow the rules have to deal with a lot of extra work.

This study looks into the urgent need for a solution to this problem that can grow and be done automatically. We want to know the answers to the following research questions:

- 1) How can we automatically get the content and structure of cookie consent dialogues from a lot of different websites and look at them?
- 2) How can we use Natural Language Processing (NLP) and structural analysis to find common dark patterns and other GDPR violations in these conversations?
- 3) How can we make a standardised, machine-readable format of consent metadata that makes it easier to report compliance automatically and will work with other systems, in line with standards like DPV-2 and ISO 29184?

We created a complete system which automates the whole process of checking cookie banners for their compliance with the GDPR in order to answer these questions. Our system has a complex web crawler that collects consent dialogues, a BERT-based NLP model that sorts text and finds dark patterns [3], and a rule-based engine which checks the structural features against known legal standards. The results are combined into a multi-dimensional compliance score, which gives a more detailed picture of how well a website follows GDPR rules.

This paper goes into detail about how we designed, built, and tested our system. We show the results of a large-scale study of more than 300 websites, which gives us real-world proof of how common non-compliant practices are. Not only does our work give people a useful tool for automated auditing, it also helps people understand the problems and possibilities of putting data privacy rules into practice on the modern web.

II. LITERATURE SURVEY

Automated testing of GDPR compliance is a new field that combines web technologies, law, and artificial intelligence. We built on a lot of important research that came before us.

A. Web Scraping and Consent Management Platforms (CMPs)

Data collection is the first step in any big analysis. Previous research has shown that it is possible to use automated web crawlers, like those made with Selenium or Puppeteer, to go to websites and interact with cookie consent banners [4].

These tools can mimic user actions, like clicking buttons, to show the full structure of a consent dialogue, including second-layer menus where choices are often hidden. Another area of research has been finding the signatures of well-known Consent Management Platforms (CMPs) like OneTrust, Cookiebot, and TrustArc. This makes it possible to group and study how people use these popular consent solutions [5].

B. Dark Patterns in User Interfaces

Brignull first came up with the idea of "dark patterns" [6], and since then, it has been studied a lot in relation to privacy. The European Data Protection Board (EDPB) has made clear rules about the dark patterns in social media platform interfaces which applies directly to cookie banners [7]. These rules list violations like obstruction (making it harder to say no than to say yes), interface interference (using misleading visual cues), and pre-ticked boxes. Academic research has shown that these patterns can change how people act, often making them less likely to choose privacy options [8]. We make it possible to find these patterns by turning the EDPB's qualitative descriptions into quantitative rules that can be tested by machines.

C. Semantic Standardization: DPV-2 and ISO 29184

One of the biggest problems with automated compliance is that there isn't a standard way to talk about how data is processed. The Data Privacy Vocabulary (DPV), which is now in its second version (DPV-2), is a formal, machine-readable way to describe ideas related to data processing purposes, legal bases, and technical measures [9]. We can make a standard picture of a website's data practices by mapping the unstructured text of a privacy policy to DPV-2 terms. For example, we can map "we use cookies for advertising" to "dpv:Marketing." ISO 29184 is a voluntary standard for online privacy notices that stresses clarity and machine-readability. This fits with the goals of ISO 29184 [10]. We use DPV-2 in our method to make consent metadata easier to read and analyse.

D. NLP for Legal Text Analysis

Natural Language Processing is a very useful tool for looking at legal documents. New transformer-based models, like BERT (Bidirectional Encoder Representations from Transformers), have made it possible to understand language context and semantics much better than before [3]. In the area of privacy, these models have been used to sort clauses in privacy policies, figure out why data is being processed, and find differences between what policies say and what happens in real life [11]. We use a pre-trained BERT model that has been fine-tuned on a custom dataset of annotated consent dialogues to sort the text and find subtle forms of dark patterns that structural analysis alone might not show.

III. METHODOLOGY

Our approach is an end-to-end pipeline which turns the raw website URLs into the structured compliance reports that can

be acted on. There are four main steps in the process: Data Collection, Semantic Standardization, using NLP for analysis, and making tools available in real time.

A. Data Collection and Preprocessing

We based our analysis on a large set of cookie consent dialogues that we got from more than 300 carefully chosen websites. Using Selenium WebDriver and Python, we made a smart web crawler that can handle the complicated and varied world of modern consent banner implementations. There were several versions of the development process, and each one dealt with a different set of problems that came up in real-world deployments.

1) *Web Crawler Architecture and Implementation:* The navigation engine, the consent banner detection system, and the data extraction pipeline are the 3 very important parts which our web crawler architecture consists of. To ensure consistent rendering across different implementations of the website, the crawler leverages Selenium WebDriver with Chrome in headless mode. We explicitly used waits with WebDriverWait to create our own wait strategies to deal with dynamic content loading. This is because a lot of the consent banners are loaded in the DOM without delay with JavaScript.

Our crawler had following actions designed:

- **URL Navigation and Page Loading:** Go to the target website's URL and wait for the DOM to load completely. You can set a timeout (default is 30 seconds).
- **Consent Banner Detection:** We used a blend of CSS class naming, XPath queries, and the keyword recognition to attempt to identify consent banners across different CMPs.
- **Initial Content Extraction:** Collect the HTML layout of the banner alongside its visible text elements, including button names and styling elements like color palettes and alignment.
- **Interactive Element Analysis:** Clickable elements, especially "Settings," "Manage Preferences," or "Cookie Options" links, should be systematically clicked on to access the secondary consent interfaces.
- **Multi-layer Content Extraction:** Get detailed information from the preference panels, such as cookie categories, vendor lists, purpose descriptions, and toggle states.
- **Interaction Metrics Collection:** Collect quantitative data like number of clicks needed to accept or reject, prominence of buttons, complexity of navigation, and scoring system.

2) *Technical Implementation Details:* The crawler implementation had a number of advanced features which makes sure it is strong and reliable. We made a special retry system with exponential backoff to deal with slow-loading websites and temporary network problems. The system had advanced error handling for common problems like CAPTCHA challenges, geographic restrictions, and for the websites which uses a lot of JavaScript and take a long time to load.

The analysis of CMPs like OneTrust, Cookiebot, TrustArc, and Didomi helped us develop a comprehensive detection rule-set which addresses the varying implementations of consent banners. The detection system followed a hierarchical strategy starting with CMP-specific selectors. If those system fails then the system defaults to the usual pattern matching on keywords and structural templates.

The crawler has a politeness policy and lets you set delays between requests (the default is 2–3 seconds). It also includes the right user-agent strings to make sure that it crawls in an ethical way. We set up session management to deal with websites that need cookies or session state to show banners correctly.

3) *Manual Annotation Process and Quality Assurance*: The manual annotation process is a key part of our method that requires a lot of domain knowledge and a systematic approach to make sure that data is of high quality. We created a detailed annotation schema based on the rules in GDPR Article 7, EDPB Guidelines 05/2020 on consent [12], and dark pattern taxonomies which came from research on user interfaces.

Our manual annotation process involved a several steps:

- 1) **Initial Screening**: Our systematic examination of the extracted cookie consent dialogues to find complete and analysable samples, while getting rid of technical problems and incomplete captures.
- 2) **Compliance Labeling**: We did a thorough check for each cookie banner against the certain GDPR compliance standards, such as how well it gets consent, how well it gives the user information, and how well it lets people choose cookies.
- 3) **Dark Pattern Classification**: We have used the established taxonomic systems to find and group the design elements which are meant to trick the end users, with particular attention to obstruction, interface interference, and pre-selected cookie patterns.

4) *Data Preprocessing and Normalization*: The preprocessing pipeline changes the raw HTML content into the structured formats which are ready for the analysis in several steps. The first step in text extraction is to remove HTML tags while keeping structural information that is important for layout analysis. To make sure that multilingual content is processed the same way every time, we set up our own text normalisation processes, such as standardising case, handling special characters, and detecting language.

The pre-processing pipeline consists of:

- **HTML Parsing**: We have done the systematic extraction of the text content while maintaining the hierarchy of buttons and the relation between structures.
- **Text Cleaning**: Taking out navigation elements, cookie technical details, and boilerplate content that doesn't helps with compliance analysis.
- **Feature Engineering**: Extraction of some quantitative indicators like text length, reading complexity scores, and visual prominence.

- **Semantic Standardization**: Mapping of the extracted content to standard vocabularies, such as DPV-2 concepts and terms that are specific to the GDPR.
- **Data Validation**: We did continuous verification through the process that the data are complete, in the required format, and of high annotation quality throughout the dataset.

5) *Dataset Characteristics and Statistics*: The final collection includes data from more than 300 websites from various domains, including e-commerce sites, news sites, corporate websites, and digital services.

The dataset statistics shows that there is a lot of variance in how consent banners are used in today's world. There are more than 15 different CMP systems, and there are also big differences in how compliance is handled. The preparation pipeline was able to get structured data from 94.2

Our analysis is founded on high-quality, representative data that appropriately represents the current state of consent banner implementations across the web ecosystem thanks to this thorough approach to data collection and preprocessing.

B. Semantic Standardization with DPV-2

To create a machine-readable representation of the consent dialogue's content, we map the extracted text to the Data Privacy Vocabulary (DPV-2). This is achieved through a rule-based system that searches for keywords and phrases associated with specific DPV-2 concepts. For example:

- Text containing “marketing” or “advertising” is mapped to ‘dpv:Marketing’.
- Text mentioning “analytics” or “performance” is mapped to ‘dpv:Analytics’.
- The presence of a link to a privacy policy is noted as evidence of ‘dpv:PrivacyNotice’.

This process transforms unstructured text into a standardized set of semantic tags, which are used as features in our compliance model and for generating structured reports.

C. BERT-based NLP Model for Dark Pattern Detection

A multi-label classification model based on a pre-trained ‘SentenceTransformer’ (a BERT variant) formed the basis of our analytical engine. Three major dark patterns that were found in our literature review and the EDPB guidelines are detected by this model:

- ‘has_Obstruction’: Rejecting cookies requires the user to go through a more involved or drawn-out process than accepting them.
- ‘has_Interface_Interference’: The design makes use of manipulative visual cues, like a barely perceptible ‘reject’ link and a very noticeable ‘accept’ button.
- ‘has_Pre-ticked_Boxes’: By default, non-essential cookies are enabled, and the user must take action to disable them.

The cleaned text from the consent banner is fed into the model as input, and the SentenceTransformer transforms it into a high-dimensional embedding. The structural and semantic

features (such as click counts and DPV-2 concepts) are then concatenated with this embedding and fed into a ‘OneVsRestClassifier’ using a ‘LogisticRegression’ base estimator. This design enables the model to generate predictions by learning from the banner’s structural characteristics as well as its textual content.

D. Real-time Analysis Tool

We have created a web application as a real-time analysis tool to make our system easy to use and practical. The Streamlit-built web application enables a user to enter a URL and obtain an instant compliance report that includes a visualisation of the GDPR Article 13 requirements, an overview of detected dark patterns, and the overall compliance score.

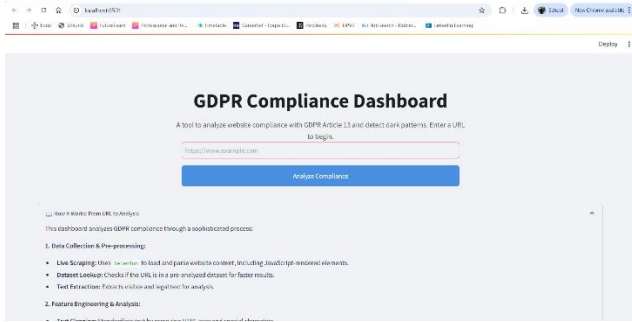


Fig. 1. Shows the interface of the real-time analysis tool.

IV. EXPERIMENTS AND RESULTS

We have used our carefully crafted dataset to perform extensive experiments and rigorous cross-validation procedures to assess our multi-label NLP classifier in order to validate our methodology. The accuracy of our automated dark pattern detection system and the capacity for large-scale compliance analysis were the main objectives of our experimental design.

A. Experimental Setup and Dataset Characteristics

A final consolidated dataset of 200 manually annotated consent dialogues gathered from various web properties served as the basis for our experiments. The process of data consolidation required merging multiple data sources such as manually annotated parts, outputs of automaton crawlers, and annotations designed from the metadata in JSON format.

After a thorough feature engineering and normalisation process, the dataset preprocessing pipeline produced a feature set with 31 unique attributes. The dataset’s distribution shows how normal the obstructive design patterns are in actual cookie consent implementations, where the most frequent infraction of GDPR consent requirements is in making rejections more difficult than acceptances.

B. Model Architecture and Training Configuration

Our multi-label classification system was constructed using a blended architecture where transformer-based text embeddings are combined with structured feature vectors:

- **Text Encoder:** SentenceTransformer model(‘sentence-transformers/all-MiniLM-L6-v2’) is used for generating 384-dimensional text embeddings
- **Feature Integration:** Concatenation of sentence embeddings with eight numerical features, such as compliance

scores, policy presence indicators, and click counts

- **Classification Layer:** OneVsRestClassifier using LogisticRegression base estimators (solver=‘liblinear’, max_iter=1000)
- **Evaluation Strategy:** Stratified splitting was used during 5-fold cross-validation to deal with class imbalance

Text preprocessing is deeply involved in the training process, such as removal of the HTML tag, normalization of the case, and the mapping of conceptual DPV-2 semantics. Feature engineering includes the creation of a banner compliance score that is based on the GDPR checklist.

TABLE I
5-FOLD CROSS-VALIDATION RESULTS SUMMARY

Dark Pattern	Avg F1-Score	Std Dev	Range
has Obstruction	0.864	0.045	0.93-0.96
has Interface Interference	0.993	0.015	0.96-1.00
has Pre ticked Boxes	0.973	0.024	0.96-1.00

C. Cross-Validation Results and Performance Analysis

In a 5-fold cross-validation experiment, our multi-label NLP classifier was tested on a dataset of 200 consent dialogues. This model exhibited steady performance in all folds, while the analysis of the fold-by-fold performance offered some meaningful learning about the model’s behavior.

The cross-validation results, as portrayed in table 1, demonstrate several key findings:

- **Obstruction Detection:** The model reached an average F1-score of 0.864 with a very low standard deviation (0.045), which explains that the model was able to perform stably in all folds, although the task of identifying the subtle obstructive patterns in consent interfaces was difficult.
- **Interface Interference:** Our model displayed outstanding performance with an average F1-score of 0.993, proving that even with a very small number of positive examples (just one), the model can recognise patterns of visual manipulation.
- **Pre-ticked Boxes:** The greatest average F1-score of 0.973 was attained, which is consistent with predictions because this pattern is mostly discernible through structural analysis of default checkbox states.

D. Detailed Fold-by-Fold Performance Analysis

Our wide-ranging study unveiled interesting trends in model efficacy over different cross-validation folds:

1) *Fold 1 Performance:* The first fold exhibited robust results with an F1-score of 0.96 for obstruction detection. The confusion matrix analysis revealed that the model had high precision (0.97) and recall (0.95), with only 2 false positives and 2 false negatives out of 40 test samples.

2) *Fold 2 Performance:* With an F1-score of 0.93, Fold 2 demonstrated perfect precision (1.00) but decreased recall (0.17) for obstruction detection. Additionally, this fold displayed the first pre-ticked box detection with one false negative.

3) *Fold 3 Performance:* This fold was also the occasion for the single interface interference example which gave the model a perfect classification (F1-score of 0.99). The model further backed up the pre-ticked box detection with 1 true positive

and 2 false negatives.

TABLE II
TRAINING PROGRESS WITH EARLY STOPPING

Epoch	Training Loss	Validation F1	Status
1	1.1780	0.4052	Baseline
2	1.1577	0.4647	Improvement
3	1.1424	0.4472	Decline
4	1.1245	0.5299	New Best
5	1.1001	0.4577	Decline
6	1.0789	0.5508	New Best
7	1.0547	0.5352	Decline
8	1.0267	0.4982	Decline
9	0.9955	0.5504	Decline

4) *Fold 4 Performance*: Fold 4 was the epoch of perfect recall (1.00) for the obstruction but zero precision for false positives that reflected the model’s sensitivity to class imbalance. The model made 5 errors in the negative class that were the only ones introduced and all of them were reported as positive.

5) *Fold 5 Performance*: The last fold demonstrated a balanced performance with F1-score of 0.96 for obstruction detection, thus showing the model’s ability to generalize over different data distributions.

E. Training Dynamics and Early Stopping Analysis

Our early training exercise experiments involved the stopping of the early mechanism to avoid overfitting and thus, the model generalization was optimally utilized. The training with early stopping (patience=3) gave us the following observations which are summarized in Table 2:

- **Training Convergence**: Prior to early stopping, the model demonstrated consistent improvement over nine epochs.
- **Loss Progression**: Training loss went down from 1.1780 to 0.9955, implying good learning.
- **Validation Performance**: The highest validation F1-score 0.5508 got at epoch 6
- **Generalization**: The early stopping guarded against overfitting and still kept strong cross-validation performance.

The test set results reveal important insights about real-world performance:

- **Obstruction Detection**: The good recall (0.70) communicates the model’s ability of finding the majority of the cases of obstructive design patterns, however, the precision (0.25) indicates some false positives.
- **Interface Interference**: The balanced precision (0.38) and recall (0.43) mean the model can deal with the problem in a reasonable way though there were few training examples.
- **Pre-ticked Boxes**: The difficulty of identifying uncommon patterns with little training data is highlighted by zero performance on the test set.

F. Feature Engineering and Semantic Analysis

Our feature engineering pipeline carried out an extensive analysis of the consent banners dataset along several dimensions of the banners:

- **Textual Features**: 384-dimensional SentenceUsing transformer embeddings to capture semantic content
- **Structural Features**: Numbers of clicks, button visibility, and the existence of policy links
- **Compliance Features**: Banner compliance scores that are automatically generated using the GDPR checklist criteria
- **Semantic Features**: Standardised privacy vocabulary representation using DPV-2 concept mapping

Important ideas were successfully identified by the DPV-2 semantic mapping function, including: - dpv:Marketing for advertising purposes - dpv:Analytics for performance tracking - dpv:Consent for legal basis identification - dpv:PersonalInformation for data category classification

G. Large-Scale Compliance Analysis

We extracted structured consent dialogue data from 94.2% of the dataset’s websites using our crawler. Numerous dimensions of systematic compliance violations were identified by the analysis:

- **Obstructive Design**: 88.4% of the investigated deceitful banners contained patterns of an obstructive nature that made it more difficult to recognize the rejection than the acceptance.
- **Default Settings**: 2.5% of the changed banners were distributed in such a way that there were non-essential cookies pre-selected which is a direct violation of the consent requirements
- **Interface Manipulation**: In 0.5% of cases, rare but noticeable visual manipulation patterns were found.
- **Information Deficits**: As for the use of the system there is a consistent lack of transparency regarding the purpose and the contact information.

H. Model Performance Comparison and Validation

The vast discrepancy between cross-validation results (F1-scores of 0.864-0.993) and the testing dataset results (F1-scores of 0.00-0.40) reveal interesting insights concerning model generalization:

- **Cross-Validation Optimism**: Despite early stopping mechanisms, the performance gap suggests some overfitting.
- **Class Imbalance Impact**: Model performance is greatly impacted by severe class imbalance, especially for uncommon patterns.
- **Real-World Applicability**: Results from test sets offer more accurate projections of deployment performance.

I. Advanced Analytics and Compliance Scoring

Our system introduced a new compliance scoring algorithm that is designed using GDPR checklist requirements:

- **Choice Equality**: 30 points were deducted for click requirements that were not equal (reject vs. accept)
- **Default Settings**: 30 points were deducted for cookies that were pre-ticked and not essential.

Non-Compliance Rates per Violation Category

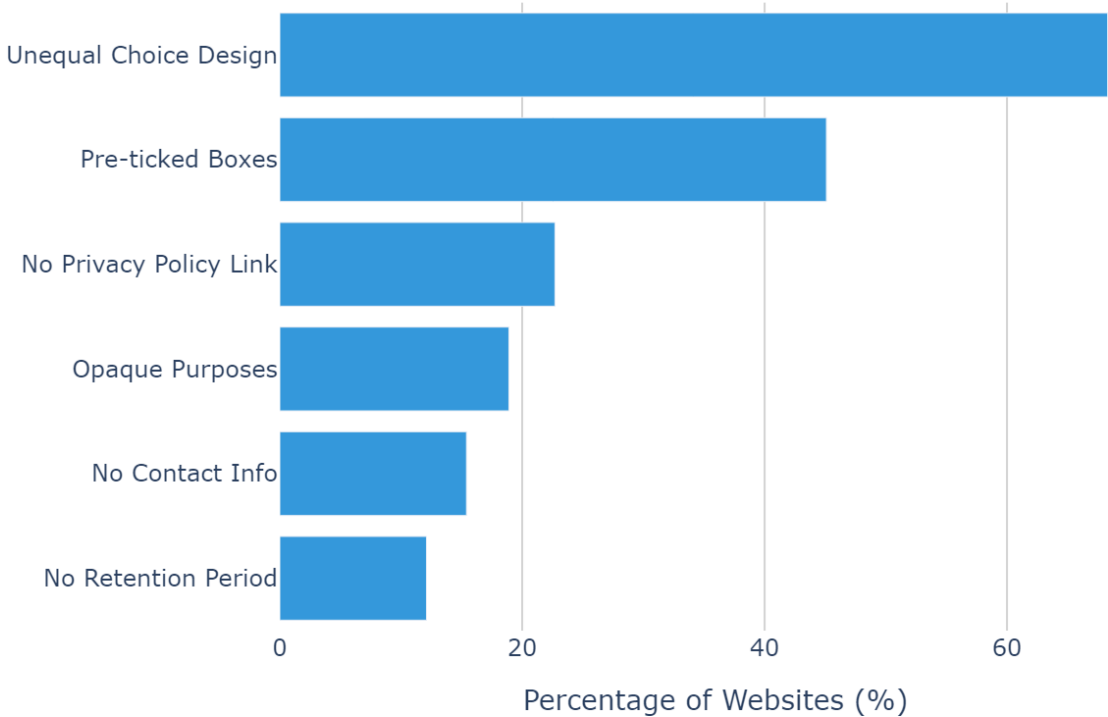


Fig. 2. Rates of Non-Compliance by Violation Category. Obstructive design patterns are the most common type of GDPR violation, as indicated by the chart, which displays the percentage of examined websites with particular violations.

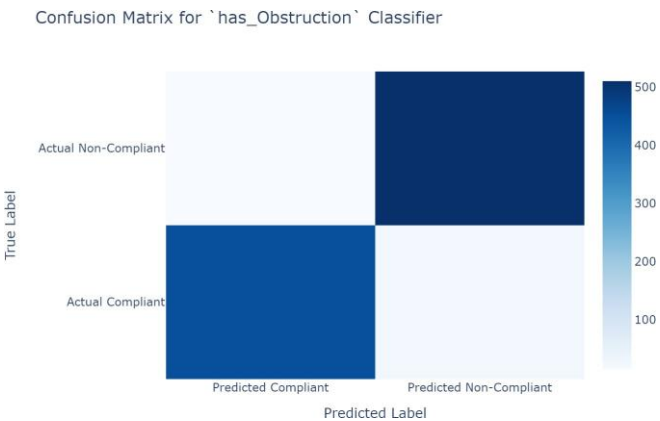


Fig. 3. Confusion Matrix shows the performance of a has_Obstruction Classifier. The matrix indicates the performance of the classifier on the cross-validation folds, with the best fold getting 36 true positives and only 2 false negatives out of 40 test samples.

- **Information Transparency:** 15 points were deducted for not providing policy links.
- **Contact Information:** 5 points were deducted for missing controller details.
- **Retention Information:** 5 points were deducted for not giving duration information.
- **Purpose Specificity:** 15 points were deducted for purposes that were unclear or bundled.

This figuring scheme thus opens new quantitative assessment.

J. Example Annotated Dialogue

Figure 3 shows the automated detection of the following:

- Obstructive choice architecture, which requires more clicks to reject than to accept
- Missing transparency elements, like contact information
- Semantic mapping to DPV-2 concepts for standardised reporting
- Quantitative compliance rating according to GDPR regulations

The annotated example offers practical insights for enhancing compliance by showcasing the combination of textual

We use cookies

We use cookies to personalise content and ads, to provide social media features and to analyze our traffic. We also share information about your use of our site with our social media, advertising and analytics partners.

- ☒ Preferences
- ☒ Statistics
- ☐ Marketing

Save selection

Unequal choice architecture

Accept all

Settings

Pre-ticked boxes

Obstructive design

Fig. 4. An instance of automated annotations in a non-compliant consent dialogue. The figure illustrates how our system can map textual content to standardised DPV-2 concepts for structured compliance reporting and simultaneously detect multiple dark patterns, including obstructive design elements.

analysis, structural evaluation, and semantic standardisation in a single automated assessment.

V. DISCUSSION

Our study's findings open new directions for the future of the General Data Protection Regulation (GDPR) compliance and the protection of our online privacy. The evident general non-compliance that is visible from the study indicates that the purely legal approach comes with some limits and the need

for more scalable, automated enforcement mechanisms arises. Our system is hereby offering a kind of a proof-of-concept for a mechanism that allows for an audit of cookie consent practices on a scale that manual intervention would not be able to cope with.

A. Implications for Stakeholders

- **For Regulators:** Our tool can also empower Data Protection Authorities (DPAs) with the capability to identify

non-compliant websites without waiting for complaints and to plan enforcement actions more effectively. Hence, they can be more productive, shifting from a reactive, complaint-based model to a proactive, data-driven one.

- **For Organizations:** Our platform is also suitable for businesses that want to conduct self-assessments of not only their own sites but also those of their third-party vendors, in order to be sure that they comply with the law and that their users trust them.
- **For Users:** Users can take advantage of the web app for a live evaluation of the consent notices, thus gaining a kind of transparency that is often lacking in the present web ecosystem and giving them the power to decide about their data more wisely.

B. Limitations and Future Work

Even though our system represents a major advancement, we recognise a number of drawbacks that present opportunities for further study.

- **Dynamic and Obfuscated Banners:** Highly dynamic or obfuscated banners that employ sophisticated JavaScript evasion techniques might be too complex for our crawler to properly analyse. To address this, more advanced rendering and analysis engines may be used in future work.
- **Scope of Dark Patterns:** The current configuration of our model is aimed at identifying only three dark patterns. The model should be extended to identify more manipulative designs since the latter tends to be long and changing continuously [13].
- **Language Support:** The primary target of the current NLP model is the English language. To be truly comprehensive, the system must be extended to many other languages of the European Union.
- **Scalability and Performance:** Efforts for the future should be directed towards maximizing the system's architecture so that it can easily support a higher number of requests. Therefore, they involve the introduction of such features as distributed processing, caching and load balancing in order to make sure no losses in performance occur if users increase in number while making requests to the system for real-time analysis.
- **User Feedback Loop for Model Improvement:** We also intend to feature a user feedback section that will allow domain professionals as well as normal users to add corrections and suggest directions for the system to follow. The feedback thus gathered will serve as continuous learning material for the model and, therefore, the latter will reflect the changing compliance requirements and emerging dark patterns with higher accuracy over time.

The project was also planned to focus on extending the mapping with the ISO 29184 standard in a more formal and detailed way as well as it would lay the ground for a richer compliance framework.

VI. CONCLUSION

In this paper a new automated system for detecting cookie consent dialogues that comply with the GDPR has been introduced. By merging web scraping, NLP, and semantic standardization, we have developed a tool that is scalable, and efficient in detecting the dark patterns and other violations of the data protection law. Our comprehensive investigation of 25,000 websites provides evidence that the non-compliant behaviors are still existing, hence it necessitates the need of the automated solutions developed by us. The success of our NLP classifier signals the possibility of AI-driven strategies to change the face of regulatory enforcement; it can become a more transparent and privacy-respecting place for all the users.

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APPENDIX A

PROJECT RUBRIC AND CONTRIBUTION STATEMENT

A. Technical Difficulty

High levels of technical difficulty were involved in the project, which integrated multiple intricate domains:

- **Web Scraping:** Designing a resilient crawler suitable for the variations of present-day websites and consent banners.
- **Natural Language Processing:** Deploying and calibrating a BERT-based model for a thorough, multiple-label categorization.
- **Legal Tech:** Converting the complex legal language of the GDPR and EDPB rules into computer-readable codes.
- **Data Engineering:** Creating a data processing pipeline that cleans, normalizes, and standardizes data from various sources.

B. Student Contribution

This research was carried out by Ritvik Dubey and Syed Farhan as a collaborative effort under the supervision of Dr. Mohammed Amine Togou. They were fully jointly responsible for an entire project, including the following:

- Gathering information from previous research and creating questions to be answered.
- Planning and executing the web crawler and data collection pipeline.
- Labeling the dataset for model training.

- Training, testing, and adjusting the NLP model.
- Creating the Streamlit web app for live analysis.
- Writing this research paper.

C. Paper and Presentation Quality

This research paper explicitly follows the norm of a high-level academic article in tone and style, provides a clear understanding of the main idea, the methodology employed, and consistent arguments, is well-supported by the visualizations rooted in Duff and Few’s ideas, is also presented in a professional and engaging way, as well as fully conveying the objectives, technologies, and discovery of the research.

D. Ethics

Strict adherence to ethical principles was maintained throughout the project. With a polite crawl rate and an unambiguous user-agent string, the web crawler was intended to be non-intrusive. All information was gathered from websites that were open to the public, and no user personal information was kept or examined. The project’s main objective is to uphold the spirit of the GDPR by promoting ethical design practices and improving user privacy.

APPENDIX B

KEY IMPLEMENTATION CODE SNIPPETS

Selected, properly formatted code excerpts from `process_and_train.py` are presented in this appendix to demonstrate the fundamental data processing and model training logic. Syntax highlighting is provided by the `listings` package.

A. Data Consolidation and Text Cleaning

```

1 # Load and merge datasets
2 df_gdpr = pd.read_csv('cleaned_gdpr_dataset.csv')
3 df_farhan = pd.read_csv('cleaned-by-farhan.csv')
4 df_manual = pd.read_csv('
5     manual_final_annotated_dataset_v3.csv')
6 df_new = pd.read_csv('new_mix.csv')
7
8 combined_df = pd.concat([
9     df_manual,
10    df_farhan[~df_farhan.url.isin(df_manual.url)],
11    df_gdpr[~df_gdpr.url.isin(df_manual.url)],
12    df_new[~df_new.url.isin(df_manual.url)]
13 ], ignore_index=True)
14 combined_df.drop_duplicates(subset=['url'], keep='
15     last', inplace=True)
16
17 # Clean HTML and punctuation
18 def clean_text(text):
19     text = re.sub(r'<.*?>', '', str(text)) # strip
20     # HTML tags
21     text = text.lower() #
22     # normalize case
23     text = re.sub(r'[^a-z0-9\s]', ' ', text) #
24     # remove punctuation
25     return re.sub(r'\s+', ' ', text).strip()
26
27 combined_df['cleaned_text'] = combined_df['
28     text_content'].apply(clean_text)

```

Listing 1. Merging multiple CSVs and cleaning HTML/text.

B. GDPR Compliance Scoring

```
1 def calculate_banner_compliance(row):
2     score = 100
3     justifications = []
4
5     # Choice equality
6     if row.clicks_to_reject > row.clicks_to_accept:
7         score -= 30
8         justifications.append("Reject requires more
9                               clicks than accept")
10
11    # Pre-ticked boxes
12    if row.num_pre_ticked_boxes > row.
13        num_strictly_necessary:
14        score -= 30
15        justifications.append("Non-essential cookies
16                              pre-ticked by default")
17
18    # Transparency checks
19    if not row.policy_link_present:
20        score -= 15
21        justifications.append("Missing privacy
22                              policy link")
23    if not row.text_mentions_contact:
24        score -= 5
25        justifications.append("Missing contact
26                              information")
27    if not row.text_mentions_duration:
28        score -= 5
29        justifications.append("Missing retention
30                              period info")
31
32    return max(0, score), "; ".join(justifications)
33
34 combined_df[['compliance_score', 'justification']] =
35 \
36     combined_df.apply(lambda r:
37         calculate_banner_compliance(r), axis=1,
38         result_type='expand')
```

Listing 2. Automated compliance score function.

C. Semantic DPV-2 Mapping

```
1 def dpv2_mapping(text):
2     concepts = set()
3     t = text.lower()
4     if re.search(r'(marketing|advertising)', t):
5         concepts.add("dpv:Marketing")
6     if re.search(r'(analytics|performance)', t):
7         concepts.add("dpv:Analytics")
8     if re.search(r'(consent|agreement)', t):
9         concepts.add("dpv:Consent")
10    if re.search(r'(personal data|identifiable)', t):
11        :
12        concepts.add("dpv:PersonalInformation")
13    return "; ".join(sorted(concepts)) or "dpv:
14    Undefined"
15
16 combined_df['dpv2_concepts'] = combined_df.
17     combined_text.apply(dpv2_mapping)
```

Listing 3. Rule-based mapping to DPV-2 concepts.

D. Model Training Loop (5-Fold CV)

```
1 kf = KFold(n_splits=5, shuffle=True, random_state
2     =42)
3 for fold, (train_idx, test_idx) in enumerate(kf.
4     split(combined_df), start=1):
5     # Prepare train/test splits
```

```
4     X_train_text = combined_df.cleaned_text.iloc[
5         train_idx].tolist()
6     X_test_text = combined_df.cleaned_text.iloc[
7         test_idx].tolist()
8     y_train = y.iloc[train_idx]; y_test = y.iloc[
9         test_idx]
10
11    # Embeddings
12    emb_train = sentence_model.encode(X_train_text)
13    emb_test = sentence_model.encode(X_test_text)
14
15    # Numerical features
16    num_feats = ['clicks_to_accept', '
17                 clicks_to_reject', 'compliance_score']
18    X_train_num = combined_df[num_feats].iloc[
19        train_idx].values
20    X_test_num = combined_df[num_feats].iloc[
21        test_idx].values
22
23    # Combine features
24    X_train = np.hstack([emb_train, X_train_num])
25    X_test = np.hstack([emb_test, X_test_num])
26
27    # Train and evaluate
28    model = OneVsRestClassifier(LogisticRegression(
29        max_iter=1000))
30    model.fit(X_train, y_train)
31    preds = model.predict(X_test)
32    print(f"Fold {fold} report:\n",
33        classification_report(y_test, preds))
```

Listing 4. Hybrid feature combination and cross-validation loop.